Cervical Cell Screening with ResNet18

Diya Shah (D25IT118) and Priya Pathak (D25IT119)
B.Tech Information Technology, CSPIT, CHARUSAT
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Abstract

Deep networks report very high accuracy on the Herlev Pap smear dataset, but a single accuracy figure says little about how a model will behave on cells it has not seen. Herlev contains only 917 single-cell images and roughly three quarters of them are abnormal, so a model that answers "abnormal" every time already scores close to 74%. A random split can also drop near-identical images into both the training and test sets, which inflates the score further.

We built a normal-versus-abnormal classifier and tried to make its score something we could defend. Exact and near-duplicate images were searched for with MD5 and difference hashes, grouped, and the data was then split by group (70/15/15, stratified on the seven original classes). No duplicates turned up in this dataset, though the pipeline is ready in case they do. Two simple baselines and a Random Forest on 11 hand-made colour and edge features served as comparisons. The main model was an ImageNet-pretrained ResNet18, fine-tuned in two phases, with two random seeds averaged.

On the 138-image validation set the ensemble reached 92.8% accuracy (95% Wilson interval 87.2 to 96.0), 92.1% sensitivity, 94.6% specificity and 98.1% AUC. The Random Forest got 87.7% accuracy and 90.9% AUC. AUC and specificity improve clearly over it; accuracy does not, since the intervals overlap at this sample size. The raw scores were a little miscalibrated (expected calibration error 0.057, down to 0.023 after Platt scaling). We also describe a three-zone triage rule that always leaves some cases with a human reader. All numbers reported here are validation results, and the test set has not been opened.

1. Introduction

Cervical cancer remains a major cause of cancer death among women. GLOBOCAN estimates for 2022 count 662,301 new cases and 348,874 deaths worldwide, and in India about 127,000 cases and 80,000 deaths, the highest death toll of any country [1]. Yet the disease is largely preventable. It develops slowly, usually after a long HPV infection, so the cell changes can be found and treated before they become invasive.

The Pap smear is the usual way of finding these changes. A technician stains cervical cells, studies them under a microscope and marks the ones that look wrong. It is slow work, it depends on who is reading, and trained cytologists are scarce in many regions. A computer-assisted first pass could help. It would not take the cytologist's place; it would quickly set aside the clearly normal cells and draw attention to the rest.

Deep learning has been applied to this problem for years. DeepPap, for instance, reported 98.3% accuracy on Herlev with convolutional networks and five-fold cross-validation [2]. We have no quarrel with such results. Our concern is narrower. Reading papers on small cytology datasets, we often could not find four things: an interval around the reported number, an account of how near-duplicate images were kept out of the test set, any handling of class imbalance beyond accuracy, and any thought about where the model output would sit in a screening workflow.

This paper therefore does not propose a new architecture. We use a standard ResNet18 and spend our effort on the evaluation. The contributions are:

1. A data pipeline that audits the dataset, removes exact and near duplicates, and splits by image group rather than by single image, with an automatic check that no group crosses a split boundary.
2. A ladder of baselines (always abnormal, random scores, Random Forest on hand-made features) showing what a ResNet18 result has to beat.
3. A full set of metrics with 95% intervals, covering both error types, plus a seed-by-seed breakdown showing that identical accuracy can hide different kinds of mistakes.
4. A calibration check and a triage rule that keeps a minimum share of cases with a human reader.

2. Background and Related Work

Herlev benchmark. The Herlev dataset was collected at Herlev University Hospital in Denmark and released as a benchmark for pattern classification [3]. Cytotechnologists labelled its 917 single-cell images into seven classes: three normal (superficial squamous, intermediate squamous, columnar) and four abnormal (mild, moderate and severe dysplasia, and carcinoma in situ). Image sizes differ.

Deep models on Herlev. Zhang et al. proposed DeepPap, a transfer-learned convolutional network, and reported 98.3% accuracy in five-fold cross-validation on the two-class task [2]. Many later studies applied pretrained CNNs to the same data, but comparing them directly is hard: the splits, class definitions, augmentation and treatment of duplicates differ or go unstated.

Residual networks and transfer learning. ResNet adds skip connections that pass a block's input on to its output, which makes deeper networks easier to train [4]. Networks pretrained on ImageNet [5] have already learned general edge and texture detectors, and reusing them is the standard choice when the target dataset is small, as it is here.

Evaluation under imbalance and leakage. About 74% of Herlev images are positive, so a constant classifier gets roughly 74% accuracy. Sensitivity and specificity separate the two kinds of error, and ROC-AUC shows how well the scores rank the cells. Balanced accuracy and the Matthews correlation coefficient adjust for imbalance. Taken together they paint a more honest picture than accuracy alone. Leakage is a separate problem: when near-identical samples sit on both sides of a split, the test score reflects memory rather than generalisation.

3. Data

3.1 Dataset and labels

We used all 917 Herlev cell images. The mask files were left out, since they only outline cell shape and are not photographs. The three normal classes hold 242 images and the four abnormal classes hold 675. The seven labels were merged into one binary target with abnormal as the positive class. For a first-look screening tool the useful question is "normal or not", and some of the seven classes would have very few images on their own. We kept the original seven labels for stratifying the split. A model that always answers "abnormal" scores 73.6% on the full set, so any classifier has to beat that number by a clear margin.

3.2 Audit

Before modelling we built a table with one row per image (path, class, binary label, size) and compared its counts with the published ones (917, 242, 675). The code prints a warning if any check fails. All checks passed and no file had to be dropped as a mask. Image sizes vary widely, from 40 x 32 up to 768 x 439 pixels.

3.3 Duplicate and near-duplicate detection

Herlev file names such as 149315671-149315740-001 and -002 suggest that several cells were cropped from the same microscope field. Such images share stain and background, and sometimes the cell itself. We did not try to prove that any two are copies, and handled similarity cautiously instead:

- **Exact copies** were found with an MD5 hash of the file bytes. Identical files were reduced to one. If identical files carried different labels, both were dropped, since the correct label cannot be known.
- **Near copies** were found with a 64-bit difference hash: the image is shrunk to 9 x 8 grey pixels and each pixel is compared with its left neighbour. Two images counted as near copies when their Hamming distance was 3 or less and their pixel dimensions matched.
- **Grouping** used union-find, so if A matches B and B matches C, all three form one group. Real families of copies are small, so any group larger than five was treated as a false match and its members were handled one by one.

The audit found 0 exact copies, 0 conflicting labels and 0 near-copy pairs. None of the 917 images had a near copy, so all 917 were kept and each formed its own group. For this dataset the group-aware split therefore behaved like an ordinary stratified split. We kept the check anyway, because the result could not be known before running it and the cost is tiny. It also means duplicate leakage does not explain the scores in this paper, at least as far as this hash and threshold can tell. Cells from the same microscope field may differ enough to pass a distance of 3, so some related images could still sit in different splits (see Section 7).

3.4 Group-aware stratified split

Groups, not single images, were assigned to train (70%), validation (15%) and test (15%). The split was stratified on the seven classes so that no split ends up short of one by chance. The script checks that no group appears in more than one split and stops if it does. Train has 641 images (472 abnormal, 169 normal) and validation has 138 (101 abnormal, 37 normal). Test also has 138 images and was not used for any decision. The share of abnormal images is similar across the three splits (73.6%, 73.2% and 73.9%).

3.5 Preprocessing

ResNet18 expects 224 x 224 inputs. Herlev crops are not square, and stretching them would distort cell shape, which is itself a diagnostic clue. Each image was therefore resized to fit inside 224 x 224 with its aspect ratio kept, and the extra border was filled with the median colour of the image edge. All images were held in memory as one array. Augmentation touched training images only: horizontal and vertical flips, rotation up to 180 degrees either way, zoom up to 10%, brightness and contrast changes up to 10%, and reflection to fill the edges. A cell has no natural "up", so none of these changes the label. Class weights came from the training set alone and were 1.896 for normal and 0.679 for abnormal.

4. Methods

4.1 Baselines

We used three reference points. The first always predicts abnormal and the second gives random scores. The third is a Random Forest of 300 trees trained on 11 hand-made features: mean and standard deviation of the red, green and blue channels; mean, standard deviation and 10th percentile of brightness; the fraction of dark pixels (a rough stand-in for nucleus size); and mean edge strength. Features were measured on the central 60% of each image so the padding could not act as a shortcut.

4.2 ResNet18 with transfer learning

ResNet18 was built in TensorFlow/Keras with the ImageNet weights loaded and the final layer removed. The new head is global average pooling (512 features), dropout at 0.3 and one sigmoid unit giving the probability of abnormal. The code checks that all 40 weighted layers received their pretrained weights and stops if not. Without that check the network could quietly train from random weights and the benefit of transfer learning would be lost.

Training ran in two phases, with binary cross-entropy, the Adam optimiser [6] and a batch size of 32.

Phase	Trainable parts	Learning rate	Epochs
1	New head only (body frozen)	1e-3	8
2	Convolutional body and head (BatchNorm frozen, model recompiled)	5e-5	up to 30

In phase 1 only the new head learns. A freshly initialised head sends back large, meaningless gradients that could damage the pretrained body. In phase 2 the body is adapted to cell images with small steps. Early stopping monitored validation AUC with a patience of 6 epochs and restored the best weights. We did not stop on accuracy because it depends on a single threshold and jumps around on a validation set this small.

One model was trained per seed (42 and 7) and their predicted probabilities were averaged. At prediction time each image was scored on four views (original, flipped left-right, flipped up-down, and both) and the four scores were averaged. The hyperparameters are common defaults judged on validation only. We ran no systematic search and do not claim they are the best possible.

4.3 Evaluation

With abnormal as the positive class we report accuracy, precision, sensitivity (recall), specificity, F1, balanced accuracy, Matthews correlation coefficient and ROC-AUC. A Type I error is a normal cell flagged as abnormal (false positive rate = 1 - specificity); a Type II error is an abnormal cell called normal (false negative rate = 1 - sensitivity). In screening the Type II error is the costlier one, because a missed case goes home with a false all-clear. The 95% intervals are Wilson score intervals [7] computed from the confusion-matrix counts. The code also produces a 1000-draw bootstrap interval [8], which gives a similar range by a different route.

4.4 Calibration and triage

Calibration was checked with a reliability plot, the Brier score and the expected calibration error (ECE), and corrected with Platt scaling [9, 10]. Triage uses two cut-offs. Scores below the low cut-off are cleared automatically, scores above the high cut-off are flagged automatically, and anything in between goes to a human reader. The cut-offs are chosen to meet three design targets: auto-clear misses at most 2% of abnormal cells, auto-flag is right at least 90% of the time, and at least 10% of all cases always stay with a human.

A threshold picked on a sample looks perfect on that same sample. To estimate how it would do on new cells we used an out-of-bag bootstrap: draw images with replacement, choose the threshold on the draw, and test it on the images left out.

5. Results

All numbers in this section come from the 138-image validation set. They are not test results.

5.1 Baselines versus ResNet18

Model	Accuracy	Sensitivity	Specificity	AUC
Always abnormal	73.2%	100%	0%	50.0%
Random scores	50.7%	48.5%	56.8%	53.1%
Random Forest (11 features)	87.7%	98.0%	59.5%	90.9%
ResNet18, seed 42	91.3%	93.1%	86.5%	98.0%
ResNet18, seed 7	91.3%	89.1%	97.3%	98.4%
ResNet18 ensemble	**92.8%**	**92.1%**	**94.6%**	**98.1%**

The ensemble also reached precision 97.9%, F1 94.9%, balanced accuracy 93.4% and an MCC of about 0.83. Its validation confusion matrix is TP = 93, FN = 8, FP = 2 and TN = 35, which gives a Type I error of 5.4% and a Type II error of 7.9%. Figure 1 shows the ROC curve and the confusion matrix.

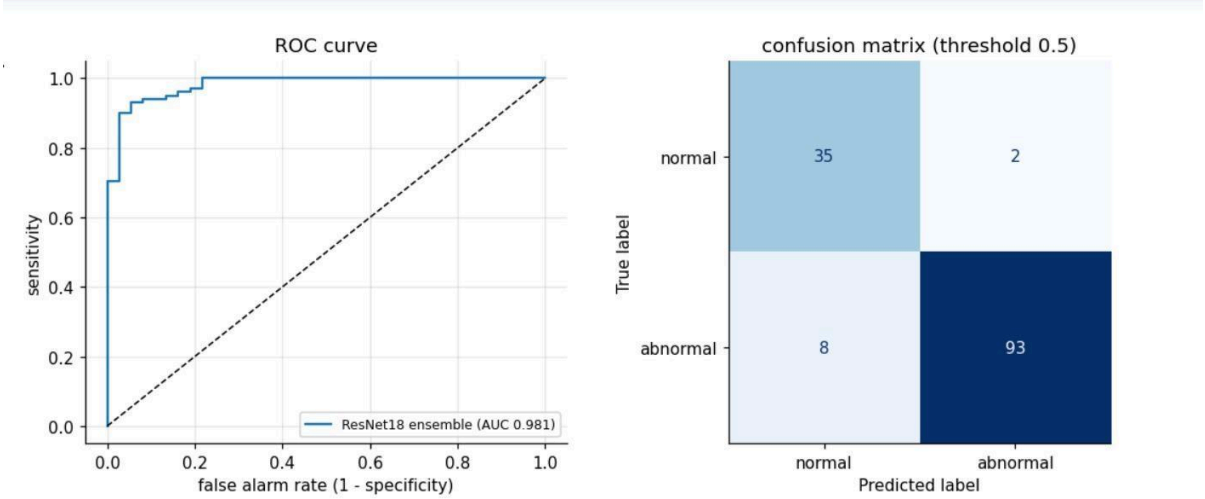


Figure 1. ROC curve of the ResNet18 ensemble (AUC 0.981), and confusion matrix at threshold 0.5. Validation set, 138 images.

The Random Forest has the higher sensitivity: it finds 99 of the 101 abnormal cells. But it clears only 22 of the 37 normal ones, because it says "abnormal" too readily. The ROC curve gives a fairer comparison. Read from our figure, the ensemble gets close to full sensitivity at about a 22% false-alarm rate, while the forest needs about 40% to reach 98% sensitivity. These values are read off a plot and are approximate.

5.2 Uncertainty

The 95% Wilson intervals for the ensemble are 87.2 to 96.0% for accuracy, 85.1 to 95.9% for sensitivity and 82.3 to 98.5% for specificity. The 1000-draw bootstrap gave close ranges: 88.4 to 97.1% for accuracy, 86.9 to 97.0% for sensitivity, 86.2 to 100% for specificity and 95.9 to 99.7% for AUC. The Random Forest accuracy interval is roughly 81 to 92%, so the two accuracy intervals overlap between about 87 and 92%. With 138 images we cannot claim that ResNet18 is better on accuracy alone. The gaps in AUC (98.1 against 90.9) and specificity (94.6 against 59.5) are larger and more convincing. Only 37 validation

images are normal, so one extra error shifts specificity by about 2.7 points, which explains the wide interval.

5.3 Same accuracy, different behaviour

Both seeds scored 91.3% accuracy, but their mistakes differed. Seed 42 missed 7 abnormal cells and raised 5 false alarms. Seed 7 missed 11 and raised only 1 (precision 98.9%). The averaged model missed 8 and raised 2. Had we picked seed 7 for its specificity, we would have kept the run that lets the most abnormal cells through, which is the wrong trade for screening. Two seeds make a small ensemble, and adding more is an easy next step.

5.4 Calibration

The raw scores had an ECE of 0.057, which fell to 0.023 after Platt scaling. The Brier score went from 0.0551 to 0.0485. In the reliability plot the raw curve sits above the diagonal in the middle range: for scores near 0.54 (6 cells in that bin) about 83% of the cells were abnormal, so the network is somewhat under-confident there. After calibration the overall error is smaller, but the curve dips below the diagonal around a score of 0.73. That bin also holds only 6 cells, and half of them were abnormal. We think this is mostly noise. The network is very sure about most images, so few validation images receive a middle score and each one moves the point a lot. Calibration changes the values of the scores but not their order, so AUC stays almost the same (0.981 before, 0.977 after). Figure 2 shows both curves.

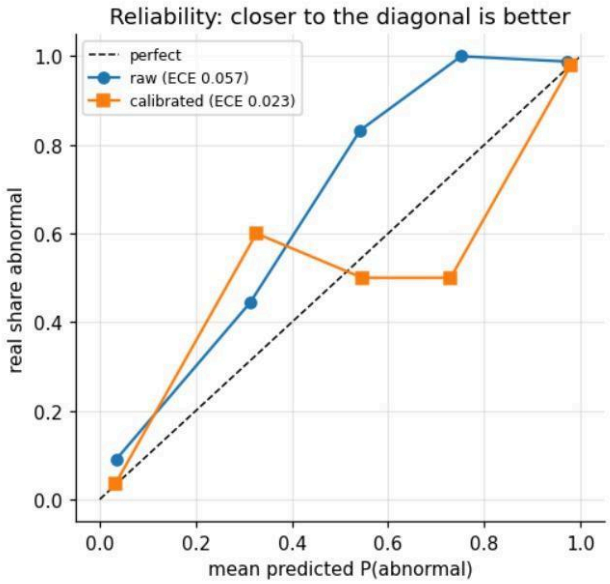


Figure 2. Reliability plot on the validation set. Raw scores (ECE 0.057) and Platt-scaled scores (ECE 0.023) against the ideal diagonal.

5.5 Triage

With the 2%, 90% and 10% targets, the cut-offs found on the validation set were 0.175 (low) and 0.553 (high). The auto-clear zone held 31 cells, 2 of them abnormal (2.0% of the 101 abnormal cells). The human-review zone held 14 cells. The auto-flag zone held 93 cells, 2 of them normal. A human would read about 10% of the cells instead of all of them. Because these cut-offs were tuned on the same

validation images, the numbers are optimistic. Values on new cells will be measured once, on the test set.

6. Discussion

The 92.8% figure is not the main thing we took away. What stood out was how much the apparent performance depended on choices that rarely appear in a results table. The majority-class baseline already scores 73%, so an accuracy in the high eighties is less impressive than it sounds. The Random Forest looks strong on sensitivity until its specificity is placed beside it. Two seeds with the same accuracy make different mistakes on the errors that count. None of this would have shown up if we had reported a single accuracy figure from a single run.

Our result is lower than the 98.3% reported for DeepPap [2], but the two cannot be compared directly. That work used five-fold cross-validation, whereas we use one group-aware split with a small validation set, and our figure is still a validation figure. We also cannot say whether earlier splits contained near-duplicates. In our own run the duplicate check found none, so leakage from exact or near copies is unlikely to explain our numbers.

For screening, a Type II error rate of 7.9% (8 missed abnormal cells out of 101) is too high for a stand-alone decision. That is the reason for the triage rule, in which uncertain scores go to a human and a minimum share of cases is never automated.

7. Limitations

- **Validation, not test.** The settings and early stopping were chosen on the same 138 images we report, so the figures may be a little optimistic. The final number has to come from one run on the untouched test set.
- **Small and narrow data.** The data are 917 images from a single hospital, with only 37 normal images in validation and two seeds.
- **Single-cell images.** A real system must find and classify cells on whole slides, across different cameras and stains. This is a research prototype.
- **Duplicate detection.** The Hamming cut-off of 3 and the group-size limit of 5 are cautious choices that we did not tune. The check found no near copies, but cells cropped from the same field may look alike without passing the hash test, so some related images may still be spread across splits.
- **No hyperparameter search.** The settings are reasonable defaults and not proven best.
- **Binary task.** Seven-class prediction and grading are not covered.

8. Conclusion and Future Work

A pretrained ResNet18, fine-tuned in two phases, separated normal from abnormal Herlev cells better than a Random Forest on hand-made features. The improvement is clear in AUC and specificity, less so in accuracy. With a leakage-controlled split, intervals and both error types reported, readers can judge how much weight the numbers deserve. Next we plan to (1) run the test set once and report that figure with

its interval, (2) check the triage cut-offs on the test set, (3) evaluate on a second cervical cytology dataset, (4) try seven-class prediction, (5) add Grad-CAM [11] to see whether the network looks at the nucleus rather than the background, and (6) test robustness to blur and colour changes.

Reproducibility note

The code runs in Google Colab with TensorFlow 2.21.0 (Keras). Random seeds 42 and 7 are fixed and the split is seeded. The pipeline also checks the dataset counts and that no group crosses a split.

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